

LC-3 Assembly Language Quiz

Write a short LC-3 assembly code snippet (1–4 lines) that satisfies each prompt. Assume any needed labels (e.g., `LOOP`, `DONE`, `MSG`) already exist in memory.

1. Write a single instruction to add the values in R3 and R4, storing the result in R2.

2. Write a single instruction to decrement R5 by 1.

3. Write a single instruction to perform a bitwise AND of R3 and R4, storing the result in R2.

4. Write a single instruction to clear register R0 to zero.

5. Write a single instruction to compute the bitwise complement of R2 and store it in R4.

6. A loop counter in R1 is decremented each iteration. Write a branch instruction that exits the loop by jumping to `DONE` when the counter reaches zero.

7. Write a branch instruction that jumps back to `LOOP` as long as R1 is not zero (positive or negative).

8. Write the instruction to prompt the user for a character and return it in R0.

9. Write a single instruction to load the value stored at memory label `NUM` into R1.

10. The memory location `PTR` holds an address. Write a single instruction to load the value at *that address* into R2.

11. Write a single instruction to load the value at memory address $R5 + 3$ into R0.

12. Write the two instructions needed to print the null-terminated string labeled `MSG` to the console.

13. Write a single instruction to store the value in R4 at the memory location labeled `RESULT`.

14. The memory location `ADDR_PTR` holds an address. Write a single instruction to store the value in R4 at *that address*.

15. Write a single instruction to store the value in R6 at memory address $R3 - 2$.

16. Write two instructions to obtain the address of the array `DECIMAL` then load R0 with the contents at index 0 of the address.

17. Write the three instructions required to load a value `COUNT`, decrement it and branch to `LOOP` if not zero.